# Impact of Material Variability on the Reliability of Nanosheet FETs

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***Abstract*—Gate-All-Around (GAA) Nanosheet Field-Effect Transistors (FETs) are replacing FinFETs due to the advancements in sub-5nm CMOS technology, which provide improved electrostatic control and scaling capabilities. The design and implementation of gate dielectrics are largely impacted by the increased integration issues brought about by the shift to sophisticated devices. In order to preserve gate capacitance while reducing leakage, high-k dielectrics exhibit intrinsic parameter fluctuations, such as dielectric constant, equivalent oxide thickness (EOT), interface trap densities, and fixed charges. The paper contains a direct comparison between Nanowire and single-layer Nanosheet FETs which represent primary GAA transistor designs for advanced logic applications. TCAD modeling allows market analysis of threshold voltage and subthreshold swing as well as drive current and leakage behavior. The superior electrostatic control comes from Nanowires but Nanosheets deliver more drive current in addition to design flexibility advantages. The research presents information that demonstrates the performance versus gate control trade-offs to help optimize future technological devices.**

***Index Terms*—Post CMOS technology, FinFET, GAAFET, 3D technology, and Nanosheet FET technology**

## I. Introduction

The advancement of CMOS technology through the past decades occurred because of an unremitting push to scale devices in accordance with Moore's Law principles [1]. The aggressive scaling process has brought about exponential increases that lead to higher transistor density and better performance and reduced energy consumption which drives quick advancements of computing systems [2]. Planar MOSFETs formed the basic building blocks when this technological development started [3]- [4]. When critical dimension reaches less than 100 nm these conventional structures developed serious short-channel effects together with leakage currents and variability issues which hindered further size reduction [5]. Semiconductor manufacturing industry migrated toward vertically oriented transistors starting from FinFETs and proceeding through Gate-All-Around (GAA) technologies including Nanosheet FETs [6]- [8]. The new generation of advanced devices solves short-channel effects at reduced node sizes through their ability to create all-surface gate control around the channel. The Nanosheet FET stands out as a leading GAA nanotechnology option for building logic devices at and below the 5 nm scale because it offers scalable design and high drive current and adjustable channel dimensions. The adoption of Nanosheet FETs brings several benefits but creates new production and design issues that primarily affect gate dielectric selection . The gate dielectric operates between the channel and gate electrode to control the fundamental transistor characteristics which include gate capacitance and threshold voltage as well as subthreshold swing and transistor switching execution [7]. High-k dielectric materials [9]- [10] including $HfO_2$ are now necessary to support high gate capacitance because of continuously diminishing device dimensions. High-k dielectrics in devices result in new variability aspects [23] which stem from dielectric constant fluctuations as well as equivalent oxide thickness (EOT) variations and interface trap densities together with fixed charge changes [24]. The variations cause important modifications in transistor behavior which degrades performance through greater power use while compromising reliability. Due to their increased sensitivity to such material inconsistencies, nanoscale devices need a better understanding of how dielectric differences affect performance parameters [25].

This paper's research uses electrostatic profiles and an evaluation of operational performance to show how gate dielectric variability impacts Nanosheet FETs [11]- [12]. The study investigates threshold voltage, on/off current ratios, and leakage characteristics at various dielectric conditions using sophisticated TCAD simulations . In order to create design methods that lessen their effects at the process level, the evaluation will identify the primary parameters that are sensitive to dielectric variability [13]. The study advances the development of Nanosheet FETs with strong performance attributes that

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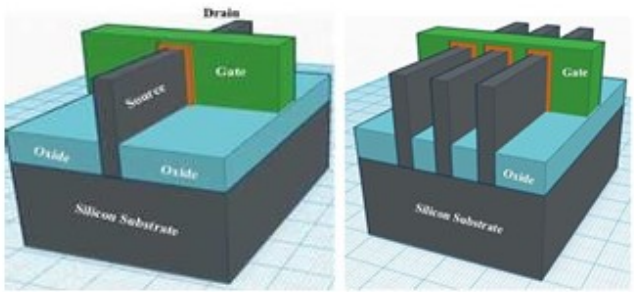


Fig. 1. Strucure of FinFET for single and Multi-FinFET [1]

meet advanced technological requirements.

## II. FINFET TECHNOLOGY AND PROGRES

Switching from 22nm to 16nm nodes resulted in an important technological breakthrough leading to the introduction of 3D integration both useful for RF and mm wave applicaions and circuit design applications [27]. This induced a shift from planar to non planar technology, i.e., 2D to 3D technology.semiconductor manufacturer have achieved a switching from conventional NanoMOSFET to FinFET transistors. FinFETs have several advantages compared to planar MOSFETs. They have better drive current capabilities, faster switching frequencies and more convenient scalability. The structure of both single and Multi-FinFET is shown in fig1 The first FinFET node has been developed by Intel and was of 22nm and FinFET device technology led to downsizing up to few nanometers. However, this downsizing reached a physical limit, again, requiring a technology change to other devices [28]. A FinFET should obey the geometrical law given by:

$$W_C = 2 \cdot H_{\text{Fin}} + W_{\text{Fin}}$$

Where $W_C$ represents the channel width, $H_{\text{Fin}}$ denotes the fin height, and $W_{\text{Fin}}$ corresponds to the fin width.

## III. NANOWIRE TECHNOLOGY AND PROGRESS

The Nanowire Field-Effect Transistor (NW-FET) operates through a cylindrical or square-shaped semiconductor nanowire channel which represents the path for current flow in Gate-All-Around (GAA) device architecture. The gate electrode creates a complete enclosure of the nanowire channel thus enabling outstanding levels of electrostatic control [18]. The entire gate encirclement in NW-FET structures prevents short-channel effects and increases subthreshold swing capability and leakage protection thereby making these devices ideal for modern low-scale nodes. Nanowires maintain dimensions between few nanometers and production techniques determine their vertical or horizontal orientation.

provides deails about Device Sructure and Doping Profile of Nanowire FET used in this design .NW-FETs provide exceptional control of potential distribution across the channel but suffer from reduced drive current levels [15] due to their small physical size which presents obstacles in high-speed operations. NW-FETs serve as essential development platforms for the next generation Nanosheet FETs and perform key roles in exploration of emerging technologies.

By using above device parametes and Doping profile , the obained strucure of Nanowire FET is shown in the fig2 .

TABLE I
DEVICE PARAMETERS AND DOPING PROFILE OF NANOWIRE [29]

| **Structure Parameters** | **Value (nm)** |
|---|---|
| Gate Length ($L_G$) | 160 |
| Nanowire Diameter ($D_{ch}$) | 10 |
| Gate Oxide Thickness | 4 |
| S/D Extension Length | 20 |
| Buried Oxide Thickness | 145 |
| **Doping Concentration** | **[cm$^{-3}$]** |
| Nanowire / Source / Drain (As) | $1.3 \times 10^{19}$ |
| Gate Poly (B) | $1 \times 10^{20}$ |

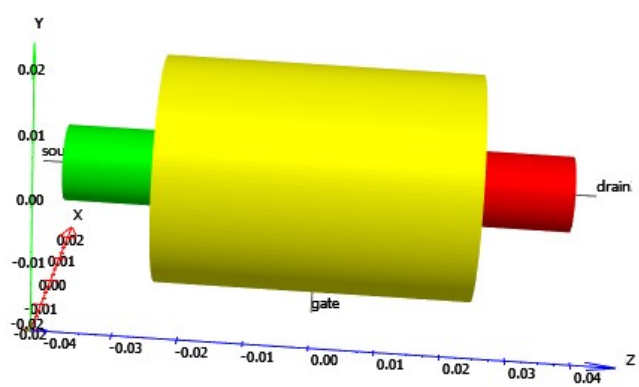


Fig. 2. Nanowire structure

## IV. NANOSHEET FET TECHNOLOGY AND PROGRESS

NS-FET technology represents an advanced gate-all-around transistor structure that extends CMOS scaling into FinFET dimensions below 5 nm nodes. NS-FET stands as a Gate-All-Around FET device which consists of horizontally stacked flat semiconductor nanosheets that receive full gate coverage. This technology provides better gate control together with less short-channel [16]- [17] issues and higher drive current than FinFETs because of its expanded effective channel width combined with complete gate containment. Performance adjustment through nanosheet width [19] modifications together with the selection of nanosheet counts enables optimal area-performace-power selections.Following its original publication, NS-FET technology saw rapid developments in both academic and industry environments. By creating small 20-nanometer gate-length devices [20] using extreme ultraviolet

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(EUV) lithography with high-k/metal gate stacks and selective epitaxy, major semiconductor companies have demonstrated functional nanosheet devices. By using nanosheet FETs, future complementary FET (CFET) devices use stacked vertical n-type and p-type devices to minimize their total footprint [21]. When constructing next-generation integrated circuits, the significant breakthrough in transistor development brought about by the NS-FETs, the FinFETs' successor technology, allows for both further downsizing and increased operating efficiency.

TABLE II
DEVICE PARAMETERS AND DOPING PROFILE OF NAOSHEET FET [29]

| **Structure Parameters** | **Value (nm)** |
|---|---|
| Gate Length ($L_G$) | 12 |
| Nanosheet Thickness ($T_{ch}$) | 5 |
| Nanosheet Width ($W_G$) | 25 |
| Spacer Length ($L_{SP}$) | 10 |
| Effective Oxide Thickness (EOT) | 1.2 |
| **Doping Concentration** | **[cm$^{-3}$]** |
| Channel | $1 \times 10^{19}$ |
| Source / Drain | $1 \times 10^{20}$ |
| Gate Poly (p-type) | $1 \times 10^{20}$ |

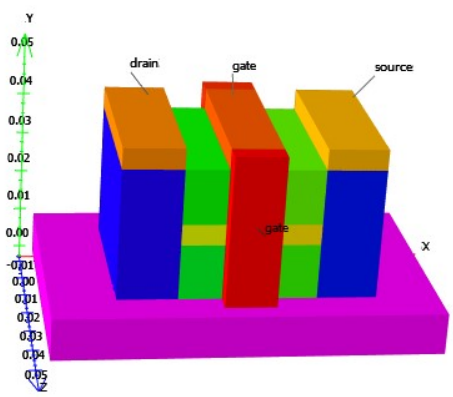


Fig. 3. Nanosheet FET - 3D structure

## A. Comparison of Different Dielectrics

A research analysis of nanosheet FETs incorporated $Al_2O_3$, $HfO_2$ and $ZrO_2$ as gate dielectrics, which produced distinct performance results depending on the material choice . From we observe that $ZrO_2$-based device achieved the maximum drain current ($I_D$) across all applied gate voltages, while $HfO_2$ and $Al_2O_3$ ranked following $ZrO_2$. The gate capacitance reaches its maximum possible level when $ZrO_2$ serves as the dielectric material because $ZrO_2$ demonstrates the highest value of dielectric constant ($\kappa \approx 25$). When gate capacitance increases, it enhances control of the channel, thereby allowing the accumulation of more carriers under steady gate voltage, which raises drive current values. The use of $ZrO_2$ as a dielectric material brought about improved switching efficiency because it produced both a reduced threshold voltage and a steeper subthreshold slope. The gate modulation capabilities of $Al_2O_3$, with its dielectric constant value of approximately 9, were inferior and produced the lowest measured $I_D$, thus making it unsuitable for high-speed or high-performance circuits. Nanosheet FET performance in next-generation CMOS technologies needs high-$\kappa$ materials for gate dielectrics to obtain optimal operational results.

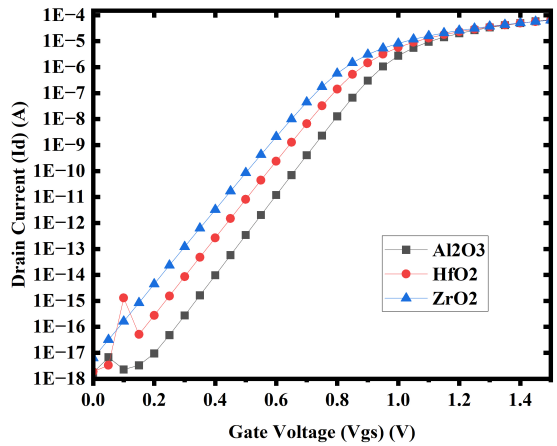


Fig. 4. $I_D$ vs $V_{gs}$ of different Dielectrics

## B. Transconductance Response of Nanosheet FET using $ZrO_2$ as Dielectric

The presented graph demonstrates how the transconductance ($g_m$) of a nanosheet field-effect transistor (NS-FET) of $ZrO_2$ as dielectric varies with changes in gate-to-source voltage ($V_{gs}$). Transconductance serves as a key performance indicator in FETs, quantifying the sensitivity of the drain current ($I_D$) to variations in gate voltage. It is mathematically defined by the differential relation:

$$g_m = \frac{dI_D}{dV_{gs}}$$

From the plot, it is evident that $g_m$ remains negligible for low values of $V_{gs}$, specifically below approximately 0.8. This behavior corresponds to the subthreshold region of operation, where the gate voltage is insufficient to induce significant charge carriers in the channel. As $V_{gs}$ increases and surpasses the threshold voltage, a sharp rise in $g_m$ is observed, indicating enhanced charge carrier injection and stronger inversion at the channel surface.

The transconductance continues to increase steadily beyond $V_{gs} \approx 1.0$, reflecting improved gate efficiency and mobility of charge carriers within the channel. High $g_m$ values are particularly desirable in high-speed and analog circuit applications, as they directly translate to better signal amplification

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capabilities.Furthermore, the consistent upward trajectory of the $g_m$ curve indicates steady device features with negligible scattering or parasitic disturbances.

This behavior underscores the essential function of gate control. In nanosheet FETs, the demand for material and structural optimization is affirmed to attain higher transconductance, which is crucial for next-generation low-power and high-performance CMOS applications.

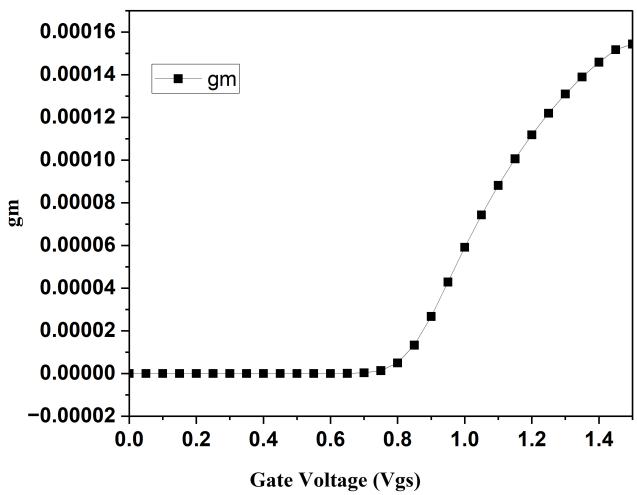


Fig. 5. $g_m$ vs $V_{gs}$ graph of Nanosheet FET using $ZrO_2$ as dielectric

### *C. Analysing Linearity parameter of Nanosheet FET using $ZrO_2$ as Dielectric*

The $g_m^2$–$V_{gs}$ curve illustrates the second order transconductance $g_{m2}$ for a nanosheet field effect transistor (NSFET). The second-order transconductance is really a measure of nonlinearity, which makes it useful in analog and RF circuit applications in big measures since it determines the first-order transconductance ‘s nonlinearity or the rate of change of the first order transconductance. It is mathematically defined as:

$$g_{m2} = \frac{d^2 I_D}{dV_{gs}^2}$$

From the graph, it can be seen that $g_{m2}$ the value remains very close to zero at low values of $V_{gs}$ which indicates that the channel formation is negligible and thus the circuit is in subthreshold region. Thus, as $V_{gs}$ increases and the device moves closer to strong inversion, $g_{m2}$ increases to a peak. This peak indicates the extreme nonlinearity in the $I_D$–$V_{gs}$ characteristic by suggesting that the increase in the drain current is expected to be much smaller for the next incremental gates voltage change. From this point onwards, $g_{m2}$ starts to decrease which implies a decrease in curvature and conduction characteristics in the saturation region.

The observation of a $g_{m2}$ peak at around the threshold transition indicates maximum nonlinearity, and hence, is relevant informing the designs for linear amplifiers. The $g_{m2}$ curve is symmetrical as per the complementary devices with predictable and controllable nonlinearity in $ZrO_2$ gate dielectrics. This is beneficial for applying techniques that linearize the behavior of circuits such as pre-distortion or negative feedback in RF front-end systems.

$HfO_2$ and $ZrO_2$, with their high-$\kappa$ nature improve the gate control and the carriers' mobility. This improves $g_m$ for higher gain but simultaneously increases $g_{m2}$, which also has a peak and needs to be fine-tuned for high linearity.

Therefore, the study of $g_{m2}$ geometry provide further insight into the linearity characteristic equations of the nanosheet FETs. In the case of the gate dielectric of $ZrO_2$, the device shows decent performance with moderate non-idealities that can be well controlled for applications in digital and RF/Analog CMOS devices.

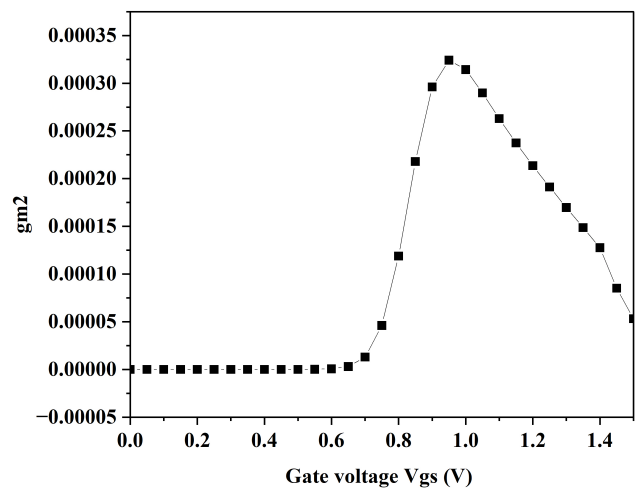


Fig. 6. Graph of $g_m^2$–$V_{gs}$ of Nanosheet FET using $ZrO_2$ as dielectric

## V. Comparison between Nanosheet and Nanowire

Naturally, in order to compare the junctionless NW FET with the NS FET on equal grounds, one has to make sure that the devices in question have the same cross-section. This is convenient in providing a standard basis for measuring basic performance metrics like on-current, and off-current. The doping profile in a channel width also has a large impact on the electrical properties of the device structures [22]. In the case of junctionless nanowire FETs doping profile is uniform round the cross section of the cylinder like structure while in case of junctionless nanosheet FETs doping profile is constructed in region of planar channel dimension. Using cross-sectional samples with consistent geometrics means that observed differences in electrical properties can be relied upon to be a function of cross-sectional geometry exclusively.

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TABLE III
COMPARING THE PARAMETERS OF NANOWIRE AND NANOSHEET FET

| Parameters | Nanowire | Nanosheet FET |
|---|---|---|
| $I_{\text{on}}$ | $1.49 \times 10^{-6}$ A | $6.61 \times 10^{-6}$ A |
| $I_{\text{off}}$ | $1.16 \times 10^{-15}$ A | $2.52 \times 10^{-18}$ A |
| $I_{\text{on}}/I_{\text{off}}$ | $1.2 \times 10^{10}$ | $2.6 \times 10^{12}$ |
| $V_{\text{th}}$ | - | 0.76 |

## RESULTS

Out of all the gate dielectrics studied, the nanosheet FETs using $ZrO_2$ showed the highest ID at all the gate voltage values, thus pointing towards a better gate control. $HfO_2$-based devices had fairly good results whereby the dielectric strength was high, but leakage current was somewhat acceptable, while $Al_2O_3$-based devices had the smallest drain current due to the low dielectric constant and poor channel control. The results revealed that the steepest subthreshold slope is in $ZrO_2$ device followed by $HfO_2$ and $Al_2O_3$ that is due to better switching efficiency with higher-*k* materials. Also, the lowest threshold voltage was obtained for $ZrO_2$, which implies an easier turn-on of the device and improved electrostatic control. Thus, dependent on what gate dielectric material was used, the characteristics of nanosheet FETs including drain current, threshold voltage, and switching functionalities are impacted.. High-*k* dielectrics like $ZrO_2$ and $HfO_2$ increase gate capacitance, thereby strengthening channel control and improving current drive and switching speed. Overall, $ZrO_2$ emerged as the most promising gate dielectric among the materials studied, making it highly suitable for advanced, low-power CMOS applications.

## CONCLUSION

Rising CMOS technology below 5nm scale chose Nanosheet FETs as the preferred successor to FinFETs since they provide enhanced electrostatic control alongside adjustable channel width features and support high-density integration capability. The research has studied fundamental Nanosheet FET properties specifically dielectric effects as well as performance measurement against Nanowire FETs. The device metrics threshold voltage, subthreshold swing and leakage current strongly depend on properties of the dielectric like thickness uniformity and material constant and interface quality. Nanosheet FET performance and reliability strongly depend on parameters stability because the complete gate provides wrap-around coverage of multiple channel layers.

The investigation shows that Nanosheet FETs offer robust gate control mechanisms, enhanced drive current, and higher process integration capabilities and design freedom. Better material control and sophisticated process monitoring are necessary for NS-FETs due to their increased sensitivity to dielectric changes. For future technological nodes to use Nanosheet FETs, careful control of dielectric variability is necessary for the best possible performance delivery. Continuous dielectric engineering research and process optimization studies are necessary for the dependable application of nanosheet-based technologies in upcoming semiconductor devices.

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